Ritesh Thawkar

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Ritesh Thawkar | Ritesh Thawkar | Google Scholar | Hugging Face

Research Profile

Researcher focused on LLMs/VLMs, agentic RAG, and multimodal systems, with experience deploying real-world LLM applications and conducting research in self-evolving LLMs/VLMs and unified understanding-generation models.

Education

- **Mohamed Bin Zayed University of Artificial Intelligence** — *Aug 2024 - Present*
  Master of Science in Computer Vision — Abu Dhabi, UAE
  - GPA: 3.55/4.00
- **Dr. Babasaheb Ambedkar Technological University** — *Feb 2021 - May 2024*
  Bachelor of Technology in Computer Science and Engineering — Lonere, India
  - GPA: 8.51/10.00

Publications

C = Conference, S = Preprint

[S1] Wafa Al Ghallabi, **Ritesh Thawkar**, Sara Ghaboura, Omkar Thawakar, Numan Saeed, Dana Al Nuaimi, Ajnas Alkatheeri, Salman Khan, Fahad Shahbaz Khan. (2026). **How Good are Foundation Models in Longitudinal MRI Disease Progression Reasoning?** *arXiv preprint*.

[S2] Shravan Venkatraman, Omkar Thawakar, **Ritesh Thawkar**, Abdelrahman Shaker, Rao Muhammad Anwer. (2026). **Perception Before Supervision: Self-Contained Visual Distillation from Counterfactual Blind Spots.** *arXiv preprint*.

[S3] **Ritesh Thawkar**, Shravan Venkatraman, Omkar Thawakar, Abdelrahman Shaker, Fahad Khan, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. (2026). **Ask, Solve, Generate: Self-Evolving Unified Multimodal Understanding and Generation via Self-Consistency Rewards.** *arXiv preprint*.

[S4] Shravan Venkatraman, **Ritesh Thawkar**, Omkar Thawakar, Rao Muhammad Anwer, Hisham Cholakkal, Salman Khan, Fahad Khan. (2026). **Paying More Attention to Visual Tokens in Self-Evolving Large Multimodal Models.** *arXiv preprint*.

[C1] Wafa Al Ghallabi, Muhammad Zaigham Zaheer, **Ritesh Thawkar**, Omkar Thawakar, Salman Khan, Fahad Shahbaz Khan. (2026). **A Multi-Agent Diffusion Approach for MRI Anomaly Segmentation via Modality-Specific LoRA Specialization.** In *WACV 2026*.

[S5] Abdelrahman Shaker, Ahmed Heakl, Jaseel Muhammad, **Ritesh Thawkar**, Omkar Thawakar, Senmao Li, Hisham Cholakkal, Ian Reid, Eric P. Xing, Salman Khan, Fahad Shahbaz Khan. (2026). **Mobile-O: Unified Multimodal Understanding and Generation on Mobile Device.** *arXiv preprint*.

[S6] Omkar Thawakar, Shravan Venkatraman, **Ritesh Thawkar**, Abdelrahman Shaker, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Fahad Khan. (2025). **EvoLMM: Self-Evolving Large Multimodal Models with Continuous Rewards.** *arXiv preprint*.

[S7] Dinura Dissanayake, Ahmed Heakl, Omkar Thawakar, Noor Ahsan, **Ritesh Thawkar**, Ketan More, Jean Lahoud, Rao Anwer, Hisham Cholakkal, Ivan Laptev, Fahad Shahbaz Khan, Salman Khan. (2025). **How Good are Foundation Models in Step-by-Step Embodied Reasoning?** *arXiv preprint*.

[C2] Omkar Thawakar, Dmitry Demidov, **Ritesh Thawkar**, Rao Muhammad Anwer, Mubarak Shah, Fahad Shahbaz Khan, Salman Khan. (2025). **Beyond Simple Edits: Composed Video Retrieval with Dense Modifications.** In *ICCV 2025*.

[C3] Wafa Al Ghallabi, **Ritesh Thawkar**, Sara Ghaboura, Ketan Pravin More, Omkar Thawakar, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. (2025). **Fann or Flop: A Multigenre, Multiera Benchmark for Arabic Poetry Understanding in LLMs.** In *EMNLP 2025*.

- **[S8]** Ayesha Ishaq, Jean Lahoud, Ketan More, Omkar Thawakar, **Ritesh Thawkar**, Dinura Dissanayake, Noor Ahsan, Yuhao Li, Fahad Shahbaz Khan, Hisham Cholakkal, Ivan Laptev, Rao Muhammad Anwer, Salman Khan. (2025). **DriveLMM-o1: A Step-by-Step Reasoning Dataset and Large Multimodal Model for Driving Scenario Understanding.** *arXiv preprint*.
- **[C4]** Sara Ghaboura, Ketan Pravin More, **Ritesh Thawkar**, Wafa Al Ghallabi, Omkar Thawakar, Fahad Shahbaz Khan, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. (2025). **Time Travel: A Comprehensive Benchmark to Evaluate LMMs on Historical and Cultural Artifacts.** In *Findings of ACL 2025*.
- **[C5]** Omkar Thawakar, Dinura Dissanayake, Ketan Pravin More, **Ritesh Thawkar**, et al. (2025). **LlamaV-o1: Rethinking Step-by-step Visual Reasoning in LLMs.** In *Findings of ACL 2025*.

Experience

- **Mohamed Bin Zayed University of Artificial Intelligence** — *Sep 2024 - Present*
  Research Assistant — Abu Dhabi, UAE
  - Developing Novel Visual Reasoning Benchmarks to stress-test the sequential logic capabilities of VLLMs.
  - Pioneering Composed Video Retrieval techniques utilizing dense video descriptions for fine-grained understanding.
  - Engineered core document retrieval systems and scalable web agents powering the enterprise Lawa.AI platform.
  - Built and optimized the AI infrastructure and retrieval backends for the Nutrigenics.care startup platform.
- **Quantiphi Analytics** — *Feb 2024 - May 2024*
  Framework Engineer Intern — Remote
  - Architected robust data-ingestion pipelines tailored for tracking small to medium business operations.
  - Developed ML-driven analytics systems to automate the extraction of actionable business insights.
  - Optimized and streamlined enterprise data processing workflows, reducing overall processing time by 20%.

Selected Projects

- **MBZUAI Web Agent Suite:** *Tools: [Agentic RAG, Web Agents, LLM Orchestration, FastAPI]*
  - Built an LLM-driven agentic pipeline that resolves MBZUAI website queries using retrieval, tool-use, and response verification.
  - Deployed the system in production, serving thousands of users and iterating prompts, tools, and routing based on real traffic.
  - Developed an interaction analytics dashboard to track user intents, content gaps, and unresolved queries for continuous Agentic RAG tuning.
- **Lawa Agent Studio:** *Tools: [LLM, RAG, Workflow Orchestration, React, FastAPI]*
  - Built a platform for creating customizable LLM-powered agentic chatbots with configurable tools, tone, and knowledge sources.
  - Implemented ingestion and indexing pipelines for website and document sources to enable fast, accurate LLM responses.
  - Added analytics and evaluation hooks to measure resolution rates, user satisfaction, and retrieval quality for LLM improvements.

- **Lawa.ai:** [🌐] *Tools: [LLM, RAG, Workflow Orchestration, FastAPI, React]*
  - Designed and developed LLM-powered web assistants to improve user interaction and engagement for websites.
  - Implemented a document retrieval system using vector databases to ground LLM responses and improve query resolution accuracy.
  - Developed backend workflows and API integration using FastAPI and websockets for scalable deployment.
- **Nutrigenics.care:** [🌐] *Tools: [LLM, Django, PostgreSQL, Recommendation System]*
  - Designed and developed the platform's web infrastructure to manage user and recipe data efficiently, enhancing scalability and usability.
  - Integrated NutritionGPT to power LLM-based recipe recommendations from image inputs and user preferences.
  - Implemented a personalized recommendation system leveraging LLMs to tailor nutrition and recipe suggestions to individual needs.
  - Optimized backend workflows using Django and PostgreSQL to ensure efficient data handling and faster response times.

SKILLS

- **AI Systems:** Agentic RAG, VLMs / MLLMs, LLM Evaluation, Vector Databases
- **Languages:** Python, TypeScript, JavaScript, C
- **ML / Deep Learning:** PyTorch, Transformers, TensorFlow, Scikit-learn, NumPy, Pandas
- **Web Technologies:** FastAPI, Flask, Django, React, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, SQLite
- **Cloud & DevOps:** AWS, Azure, Git

ADDITIONAL INFORMATION

Languages: English (Intermediate), Hindi (Advanced), Marathi (Advanced)
Interests: Reading, Music, Sports (Cricket, Badminton)